

BHASKAR PRATAP

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Education

Ajay Kumar Garg Engineering College

Bachelor of Technology in Computer Science and Engineering (CSE) – CGPA: 8.51*

Oct. 2023 – Present

Ghaziabad, Uttar Pradesh

Maria School

Intermediate (Class XII — ISC) – Percentage: 91.75%

March 2021 – June 2022

Jaspur, Uttarakhand

Maria School

High School (Class X — ICSE) – Percentage: 91.00%

March 2019 – April 2020

Jaspur, Uttarakhand

Coursework / Learnings

- Data Structures & Algorithms (Java)
- Object Oriented Programming
- Database Management Systems
- Operating Systems
- Computer Networking

Projects

Theatre: Event-Driven Seat Booking System | Java, Spring Boot, Kafka, Redis, PostgreSQL, JWT 📄 GitHub 2026

- Engineered a high-concurrency seat booking engine using **Spring Boot** and **PostgreSQL**, implementing robust **transactional atomicity** and **state management** to ensure precise reservation lifecycles and data consistency.
- Architected an asynchronous notification system with **Apache Kafka**, utilizing a **Producer-Consumer** model to broadcast real-time booking and cancellation events.
- Architected a multi-tiered caching strategy using **Redis** to optimize movie catalog retrieval and seat availability lookups by **70%**, implementing automated **cache-eviction policies** to ensure strict data synchronization with the **PostgreSQL** source of truth.
- Engineered a robust security layer with **Spring Security**, featuring **Role-Based Access Control (RBAC)** for User, Admin, and Super-Admin hierarchies.
- Demonstrated strong **Computer Science fundamentals** by resolving **race conditions** and **Unique Constraint violations** through precise JPA relationship management and leveraging the **Java Streams API** for thread-safe, high-concurrency data transformations.

PeerLink: Distributed P2P File Transfer System | Socket Programming, Multi-threading, File-Handling 📄 GitHub 2026

- Engineered a **decentralized peer-to-peer (P2P) architecture** using **Java Socket Programming** and a custom port-allocation system for direct device-to-device data exchange.
- Developed a high-concurrency backend capable of dynamically initializing **dedicated file servers** on generated ports to facilitate **isolated, high-speed data streams**.
- Architected the **core networking logic** to manage **asynchronous peer discovery** and direct **TCP/IP communication** between distributed nodes.
- Implemented **robust multi-threading** in Java to handle concurrent file requests, ensuring system stability and **resource optimization** during simultaneous data transfers.
- Designed a modular **Low-Level Design (LLD)** that decouples core file-handling logic from **transport protocols**, ensuring the system is highly extensible and scalable.
- Streamlined the development lifecycle with **Maven** for dependency management and created cross-platform **automation scripts** to manage server instances efficiently.

Data Structures And Algorithms

- Solved **700+** problems in Java, with a global rank **70k** on Leetcode.
- Actively participate in contests with consistent performance and problem-solving growth.
- Contest rating **max(1617)**-Leetocde .

Technical Skills

Languages: Java , PostgreSQL

Technologies/Frameworks: Spring Boot, Spring Security (JWT/RBAC), Kafka, Redis, Tailwind CSS.

Distributed Systems: Event-Driven Architecture, P2P Networking, Socket Programming, REST APIs , Collection Frameworks.

Developer Tools: IntelliJ IDEA, Git/GitHub, Postman, Maven, Docker, Docker Compose, VS Code.